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Unit – 4 : Session – 2 : SLO – 2 | Operations on Binary Trees — Insertion and Deletion

■ I. Fill in the Blanks

- In a binary tree, each node has at most TWO children.
- The first node inserted into a binary tree is called the ROOT node.
- In a binary tree insertion algorithm, if a node's left child is NULL, the new node is inserted at the LEFT position.
- In a level-order insertion (used for a complete binary tree), the new node is inserted at the first available position from LEFT to RIGHT.
- In recursive insertion of a binary tree, the base case occurs when the current node is NULL.
- Inserting into a binary tree without any specific order results in a GENERAL (unordered/random) binary tree.
- In a binary tree, a node with ZERO (no) children can be deleted by simply removing the node.
- A node with one child can be deleted by linking its parent directly to its CHILD.
- When deleting a node with two children, the successor or predecessor must be deleted from its ORIGINAL location.